Lars Larsen

Hardware Test, Verification and Engineering Tools

Portland, OR (remote or hybrid) · lars@lzxindustries.net · creatorlars.github.io · github.com/creatorlars · linkedin.com/in/lars-larsen-7b4a429

SUMMARY

Hardware engineer focused on test and verification: hardware-in-the-loop video test rigs, simulation models checked against RTL, oscilloscope automation, and in-house CAD and PLM tools. Led product design at LZX Industries since 2011, with a three-person core team.

EXPERIENCE

Founder & Principal Engineer, Verification & Tooling, LZX Industries

Portland, OR · 2011 – Present

Maker of video synthesizers and video processing instruments (DBA 2011, LLC 2017). Led product design in a three-person core team, working with partners who handle firmware/HDL and board layout.

- Led product design as our team took more than 100 hardware products to production across six Eurorack series, standalone instruments, DIY kits, cases and power. About 240 production circuit board designs and 140 metal front panels. $6.4M in direct sales (2018 to 2026), plus a dealer channel.
- Videomancer verification (2026): built a test harness that scores GHDL simulation output and real HDMI captures with the same SSIM/PSNR checks. Full sweep is 984 cases, with 46 regression tests added for defects found during release candidates.
- Chromagnon (2026): wrote a Python model of the instrument that matches the iCE40 RTL and GHDL bit for bit, with nine block-level test runners (all 76 injected faults caught in simulation), Rigol MSO5000 control over SCPI, and 88 reference-vector tests for the browser version.
- Simulation (2026, AI-assisted tools): ngspice reviews with derating rules and IBIS-based driver models. A 200-check review of a clock board found 9 problems, fixed before fabrication. Shared model library of about 60 parts.
- Engineering tools (2026, AI-assisted): a C++ fork of KiCad (LZX Designer) with custom file formats, Eurorack templates and an InvenTree parts library; Python scripts that generate schematics, placement, routing and silkscreen; a parametric sheet-metal case generator.
- Software: led the Shopify Hydrogen/React storefront since 2023 (about 58k lines, Playwright tests). In 2026, using AI coding agents: a Python operations CLI (about 1,000 commands, 3,500 tests) linking Shopify, InvenTree, ERPNext, JLCPCB and UPS; a Rust/Tauri firmware updater; self-hosted Forgejo CI and Authentik SSO.

Operations Manager, Darkplace Manufacturing

Portland, OR · 2015 – 2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX, 2008–2010 and 2011–2015
- Lead Service Technician, Switched On (synthesizer sales and repair), Austin TX, 2010
- New Media Developer, FUNimation Entertainment, Flower Mound TX, 2007–2008
- Multimedia Developer, TargetScope, Plano TX, 2006–2007

SELECTED PROJECTS

- **Videomancer SDK (GPL-3.0)**: iCE40 video platform, program ABI, signed package format and GHDL image tester. About 190 C++ and 350 VHDL unit tests. github.com/lzxindustries/videomancer-sdk
- **Chromagnon web simulator**: Browser version of the Chromagnon instrument model (WebGL), checked against reference vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **LZX Industries storefront**: Shopify Hydrogen/React storefront with product pages, firmware downloads and online tools. lzxindustries.net

SKILLS

Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI coding tools (Claude Code, Cursor)
Firmware & FPGA	C/C++ (C++17/20, no heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing, VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40), Vivado (Zynq-7000)
Electronics	Analog, mixed-signal and video (SD to 1080p, composite, S-Video, YPbPr, RGB, HDMI, genlock), discrete high-speed analog, power entry and regulation, ngspice/SPICE, IBIS, DipTrace, Eagle, KiCad, DFM, BOM costing, board bring-up, oscilloscope automation
Product & operations	Concept to production, PLM (MRPeasy, InvenTree), ERP, supplier and contract manufacturer management, pricing, documentation, open-source hardware
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosures, injection-molded knobs, front panel graphics, Blender rendering

EDUCATION

- B.A., Radio, Television & Film, University of North Texas, 2006